

# SmartTrack API - Driver

**Base URL:** `{{base_url}}`

Driver-facing slice of the SmartTrack API, covering both regular drivers and guest drivers: login, trip status updates, attendance marking/QR scan, lost & found reporting, and the guest-driver-only guest-trip and ticket endpoints.

Recent API changes (read before integrating): - No more "your assigned bus." Get Me and any driver lookup no longer return `assigned_bus_id` / `assigned_bus_number`; buses no longer return `driver_id` / `driver_name` either. A driver only has a bus for as long as a trip is actually running -- read it from List Trips / Take Over Trip, not from your own profile. - QR scans return less. Scanning a bus's safety QR no longer includes `driver_id`; scanning a route's QR no longer includes `bus_id` / `driver_id`. Don't pre-check "is this my bus" from the scan itself -- start the trip and let the server enforce it. - Bus-breakdown handover, now documented below. See "Take Over Trip after Breakdown" -- scan the ORIGINAL (breakdown) bus's safety QR, not the replacement bus's, and only after a school admin has named you as the standby driver. - Lost & Found auto-fill changed source. Report Lost Item without an explicit `bus_id` now falls back to your current/most recent trip's bus instead of a persistent assignment. Send `bus_id` explicitly when you already have it on screen.

How to use: 1. Run "Login - Driver" (or "Login - Guest Driver" for the guest-trip / ticket endpoints) first -- it saves the matching `driver_token` / `guest_driver_token` collection variable used to authenticate the rest of the requests in this doc. 2. `trip_id`, `student_id`, `student_qr_code` etc. are captured automatically as you exercise the flows.

Seeded demo accounts (see `backend/src/db/seed.js`): `driver@smartrack.ae`, `guest@smartrack.ae`.

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## Collection Variables

Seeded demo accounts and IDs used throughout this reference. Values left blank are captured automatically at runtime (tokens, ids, QR codes) rather than fixed ahead of time.

| Variable                        | Default / Notes                        |
|---------------------------------|----------------------------------------|
| <code>baseUrl</code>            | <code>http://localhost:5000/api</code> |
| <code>driver_email</code>       | <code>driver@smartrack.ae</code>       |
| <code>driver_password</code>    | <code>Driver@123</code>                |
| <code>guest_driver_email</code> | <code>guest@smartrack.ae</code>        |

|                       |                       |
|-----------------------|-----------------------|
| guest_driver_password | Guest@123             |
| driver_token          | (captured at runtime) |
| guest_driver_token    | (captured at runtime) |
| school_id             | GREENFIELD            |
| student_id            | (captured at runtime) |
| student_qr_code       | (captured at runtime) |
| bus_id                | (captured at runtime) |
| safety_qr_code        | (captured at runtime) |
| driver_id             | (captured at runtime) |
| trip_id               | (captured at runtime) |

## 1. Auth

### 1. Login - Driver

**POST** `{{base_url}}/auth/login`

#### Request Body

```
{
  "email": "{{driver_email}}",
  "password": "{{driver_password}}",
  "school_id": "{{school_id}}"
}
```

#### Response Example

200

```
{
  "user": {
    "id": "USR-12345",
    "name": "Jane Doe",
    "email": "user@example.com",
    "phone": "+971500000000",
    "role": "driver",
    "school_id": "SCH-001",
    "school_name": "Greenfield International",
    "avatar": "https://example.com/avatar.jpg",
    "fcm_token": "demo-device-token",
    "created_at": "2023-01-01T00:00:00.000Z",
    "last_login": "2023-01-02T08:00:00.000Z"
  },
  "token": "<jwt>"
}
```

#### Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.token) pm.collectionVariables.set('driver_token', data.token);
```

## 2. Login - Guest Driver

**POST** `{{base_url}}/auth/login`

### Request Body

```
{
  "email": "{{guest_driver_email}}",
  "password": "{{guest_driver_password}}",
  "school_id": "{{school_id}}"
}
```

### Response Example

200

```
{
  "user": {
    "id": "USR-12345",
    "name": "Jane Doe",
    "email": "user@example.com",
    "phone": "+971500000000",
    "role": "driver",
    "school_id": "SCH-001",
    "school_name": "Greenfield International",
    "avatar": "https://example.com/avatar.jpg",
    "fcm_token": "demo-device-token",
    "created_at": "2023-01-01T00:00:00.000Z",
    "last_login": "2023-01-02T08:00:00.000Z"
  },
  "token": "<jwt>"
}
```

### Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.token) pm.collectionVariables.set('guest_driver_token', data.token);
```

## 3. Get Me (driver)

**GET** `{{base_url}}/auth/me`

### Request Headers

| Header        | Value                                |
|---------------|--------------------------------------|
| Authorization | Bearer <code>{{driver_token}}</code> |

### Response Example

200

```
{
  "user": {
    "id": "USR-12345",
    "name": "Jane Doe",
    "email": "user@example.com",
    "phone": "+971500000000",
    "role": "driver",
    "school_id": "SCH-001",
    "school_name": "Greenfield International",
    "avatar": "https://example.com/avatar.jpg",
    "fcm_token": "demo-device-token",
    "created_at": "2023-01-01T00:00:00.000Z",
    "last_login": "2023-01-02T08:00:00.000Z"
  }
}
```

```
}  
}
```

#### 4. **NEW** Update FCM Token (driver) [NEW]

**PATCH** `{{base_url}}/auth/fcm-token`

New self-service endpoint - any role can register its own push token. Sending is currently stubbed server-side (logged, not actually delivered).

##### Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

##### Request Body

```
{  
  "fcm_token": "demo-device-token-abc123"  
}
```

##### Response Example

200

```
{  
  "user": {  
    "id": "USR-12345",  
    "name": "Jane Doe",  
    "email": "user@example.com",  
    "phone": "+971500000000",  
    "role": "driver",  
    "school_id": "SCH-001",  
    "school_name": "Greenfield International",  
    "avatar": "https://example.com/avatar.jpg",  
    "fcm_token": "demo-device-token",  
    "created_at": "2023-01-01T00:00:00.000Z",  
    "last_login": "2023-01-02T08:00:00.000Z"  
  }  
}
```

#### 5. Change Password (driver)

**PATCH** `{{base_url}}/auth/change-password`

Sets the new password to the same value as before, so it's safe to run repeatedly without breaking other requests.

##### Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

##### Request Body

```
{  
  "currentPassword": "{{driver_password}}",  
  "newPassword": "{{driver_password}}"  
}
```

##### Response Example

200

```
{
  "message": "Password updated successfully"
}
```

## 6. Logout (driver)

**POST** `{{base_url}}/auth/logout`

Stateless JWT - this is a 204 no-op; the token remains valid until it expires. Re-run 'Login - Driver' afterwards if you need a fresh `{{driver_token}}`.

### Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

### Response Example

204

No response body.

## 3. QR [NEW]

### 1. Resolve Student QR (driver)

**GET** `{{base_url}}/qr/{{student_qr_code}}`

Run '2. Lookups' first so `{{student_qr_code}}` is populated. Returns `{ type: 'student', entity: {...} }`.

### Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

### Response Example

200

```
{
  "type": "<student|bus|route>",
  "entity": { ... }
}
```

## 4. Trips

### 1. List Trips (driver - scoped to own trips)

**GET** `{{base_url}}/trips`

[CHANGED] Only returns trips belonging to this driver, regardless of any `driver_id` query param passed.

### Request Headers

| Header        | Value                   |
|---------------|-------------------------|
| Authorization | Bearer {{driver_token}} |

## Response Example

200

```
{
  "trips": []
}
```

## 2. Update Trip Status (driver - own trip)

**PATCH** {{base\_url}}/trips/{{trip\_id}}

A driver may only PATCH the status field of their own trip (pre-existing behavior, unchanged).

### Request Headers

| Header        | Value                   |
|---------------|-------------------------|
| Authorization | Bearer {{driver_token}} |

### Request Body

```
{
  "status": "in_progress"
}
```

## Response Example

200

```
{
  "{{trip_id}}": { ... }
}
```

## 3. **NEW** Take Over Trip after Breakdown (driver) [NEW]

**POST** {{base\_url}}/trips/take-over

[NEW] A standby driver completes a bus-breakdown handover by scanning the ORIGINAL (breakdown) bus's safety QR code -- not the replacement bus's, since that's the only one they have physical access to. Requires a school admin to have already named this driver as new\_driver\_id on the bus\_transfers record (via the admin console's Bus Transfer screen); otherwise this 403s with "You are not the assigned driver for this transfer." On success, trips.driver\_id switches to this driver and the bus\_transfers record is marked completed.

### Request Headers

| Header        | Value                   |
|---------------|-------------------------|
| Authorization | Bearer {{driver_token}} |

### Request Body

```
{
  "safety_qr_code": "{{safety_qr_code}}"
}
```

## Response Example

201

```
{
  "take-over": { ... }
}
```

## Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.trip) pm.collectionVariables.set('trip_id', data.trip.id);
```

# 5. Attendance

## 1. Mark Attendance (driver)

**POST** `{{base_url}}/attendance`

[CHANGED] Now 403s if trip\_id doesn't belong to this driver.

### Request Headers

| Header        | Value                                |
|---------------|--------------------------------------|
| Authorization | Bearer <code>{{driver_token}}</code> |

### Request Body

```
{
  "trip_id": "{{trip_id}}",
  "student_id": "{{student_id}}",
  "status": "present"
}
```

## Response Example

201

```
{
  "record": { ... }
}
```

## Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.record) pm.collectionVariables.set('attendance_id', data.record.id);
```

## 2. **NEW** Scan Attendance (driver) [NEW]

**POST** `{{base_url}}/attendance/scan`

Resolves the student by QR code and marks them present in one call, stamping pickup\_time or drop\_time depending on the trip's type.

### Request Headers

| Header        | Value                                |
|---------------|--------------------------------------|
| Authorization | Bearer <code>{{driver_token}}</code> |

## Request Body

```
{
  "trip_id": "{{trip_id}}",
  "qr_code": "{{student_qr_code}}"
}
```

## Response Example

200

```
{
  "scan": { ... }
}
```

## 3. List Attendance (driver - own trips only)

**GET** `{{base_url}}/attendance`

[CHANGED] Scoped to this driver's own trips.

## Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

## Response Example

200

```
{
  "records": []
}
```

# 7. Lost & Found

## 1. Report Lost Item (driver)

**POST** `{{base_url}}/lost-found`

## Request Headers

| Header                     | Value                                |
|----------------------------|--------------------------------------|
| <code>Authorization</code> | <code>Bearer {{driver_token}}</code> |

## Request Body

```
{
  "bus_id": "{{bus_id}}",
  "driver_id": "{{driver_id}}",
  "description": "Blue water bottle found on back seat"
}
```

## Response Example

201

```
{
  "item": { ... }
}
```

#### Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.item) pm.collectionVariables.set('lost_found_id', data.item.id);
```

## 8. Guest Trips

### 1. Create Guest Trip (guest\_driver)

**POST** `{{base_url}}/guest-trips`

[CHANGED] The name/phone above are intentionally wrong - check the response and confirm guest\_driver\_name/guest\_driver\_phone came back as the guest\_driver account's real name/phone instead, not what was sent.

#### Request Headers

| Header        | Value                                      |
|---------------|--------------------------------------------|
| Authorization | Bearer <code>{{guest_driver_token}}</code> |

#### Request Body

```
{
  "guest_driver_name": "SPOOFED NAME (should be overridden)",
  "guest_driver_phone": "000-000-0000 (should be overridden)",
  "bus_registration": "BUS-TEST-1"
}
```

#### Response Example

**201**

```
{
  "trip": { ... }
}
```

#### Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.trip) pm.collectionVariables.set('guest_trip_id', data.trip.id);
```

### 2. List My Guest Trips (guest\_driver)

**GET** `{{base_url}}/guest-trips`

[CHANGED] Scoped to guest trips matching this account's own phone number.

#### Request Headers

| Header        | Value                                      |
|---------------|--------------------------------------------|
| Authorization | Bearer <code>{{guest_driver_token}}</code> |

#### Response Example

200

```
{
  "trips": []
}
```

## 10. Tickets

### 1. Create Ticket (guest\_driver)

**POST** `{{base_url}}/tickets`

#### Request Headers

| Header        | Value                                      |
|---------------|--------------------------------------------|
| Authorization | Bearer <code>{{guest_driver_token}}</code> |

#### Request Body

```
{
  "type": "Vehicle Issue",
  "priority": "medium",
  "description": "AC not working on assigned bus"
}
```

#### Response Example

201

```
{
  "ticket": { ... }
}
```

#### Auto-Captured Variables (test script)

```
const data = pm.response.json();
if (data.ticket) pm.collectionVariables.set('ticket_id', data.ticket.id);
```

### 2. List My Tickets (guest\_driver) [NEW - mine=true]

**GET** `{{base_url}}/tickets?mine=true`

Narrows to just this caller's own reported tickets, regardless of role.

#### Request Headers

| Header        | Value                                      |
|---------------|--------------------------------------------|
| Authorization | Bearer <code>{{guest_driver_token}}</code> |

| Query Param | Example Value |
|-------------|---------------|
| mine        | true          |

#### Response Example

200

```
{  
  "tickets": []  
}
```